

Emilia Ma

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EDUCATION

University of Toronto

Sept 2024 – Expected May 2029

B.Sc. in Computer Science — Dean's List Scholar

GPA: 3.76/4.0

- Coursework: Data Structures & Algorithms, Software Design, Human Computer Interaction, Systems Programming, Computer Systems & Digital Logic, Theory of Computation, Probability & Statistics

SHAD — National STEM Enrichment Program

Summer 2023

- Proposed the team's core project concept around urban design and transit, helping shape the solution's direction and scope while collaborating on research, development, and the final presentation.

EXPERIENCE

Ontario Public Service | Backend Dev/Junior Technical Analyst

Sept 2026 – Apr 2027

- Incoming software development intern starting September 2026 in the Digital Solutions Branch.
- Contributing to Java backend development for a large scale modernization of Ontario's driver and vehicle service systems, migrating legacy government systems to a new digital platform.
- Working in a multi layered enterprise backend architecture, on core application/business logic layers.

French Tutor & English Volunteer

2021–2024

- Taught English remotely to children in China with an emphasis on real conversational practice.
- Tutored a local student in French, tailoring sessions to her learning style.

PROJECTS

Tempo | *React, TypeScript, Node.js, Express, MongoDB, Better Auth, Railway*

2026–Present

Full-stack calendar focused on making scheduling faster and more intuitive through streamlined interactions.

- Independently designed, developed, and deployed Tempo end to end, implementing account authentication and user-scoped event storage so users can access their own calendars across sessions and devices.
- Built a regex parser that extracts times as users type, combining event titles and times in one field.
- Implemented REST endpoints and MongoDB persistence for event creation, updates, and retrieval.

GirlMath | *Python, FastAPI, React Native, TypeScript, MongoDB, DigitalOcean, Gemini API*

2026

AI powered journaling app supporting girls in STEM by tracking confidence, motivation, and learning trends.

- Led backend dev in a 36-hour hackathon, built FastAPI logic and REST endpoints for a 4 person team.
- Integrated the Gemini API to derive confidence, motivation, and learning trends from journal entries.
- Built frontend API client for backend communication and deployed the FastAPI service to DigitalOcean.

Munchables | *Java, Swing, OkHttp, JSON, Spoonacular API, HTML/CSS, JUnit*

2025

Four-person Java Swing social platform created with Clean Architecture and JUnit-tested domain logic.

- Designed a reusable Swing post component used across all post views, centralizing image rendering, comments, likes, and interaction logic.
- Integrated Spoonacular API, displaying nutrition data within the application using custom HTML/CSS.
- Implemented persistence via repository interfaces, isolating domain logic from storage infrastructure.

Game Recommendation System | *Python, Streamlit, Steam Web API, JSON*

2025

Graph based Steam game recommender using user selected preference filters.

- Developed a rate limited steamAPI pipeline with local caching for fast offline graph recommendations.
- Built Streamlit UI with custom HTML/CSS for scrollable game list browsing & additional details.

SKILLS

- **Programming Languages:** Python, Java, TypeScript, JavaScript, C, HTML/CSS, MIPS Assembly
- **Technologies & Tools:** Git, MongoDB, React/React Native, Node.js, Express, FastAPI, Maven, JUnit, Expo, Railway, DigitalOcean, Java Swing, Figma, REST APIs, Streamlit
- **Languages:** Proficient in English, French, and Chinese